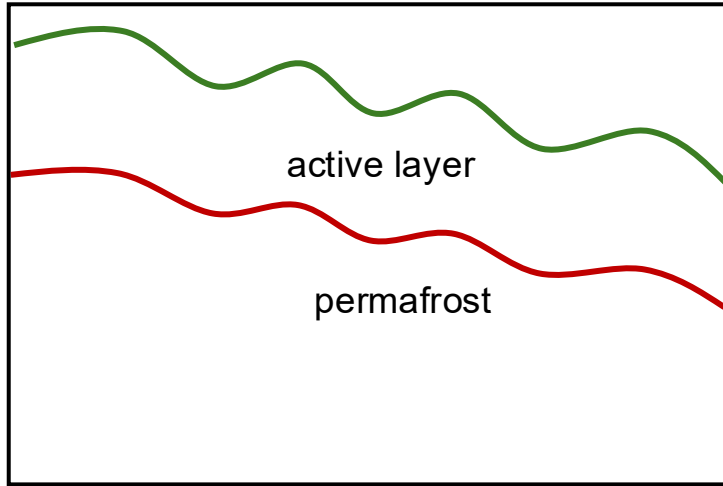
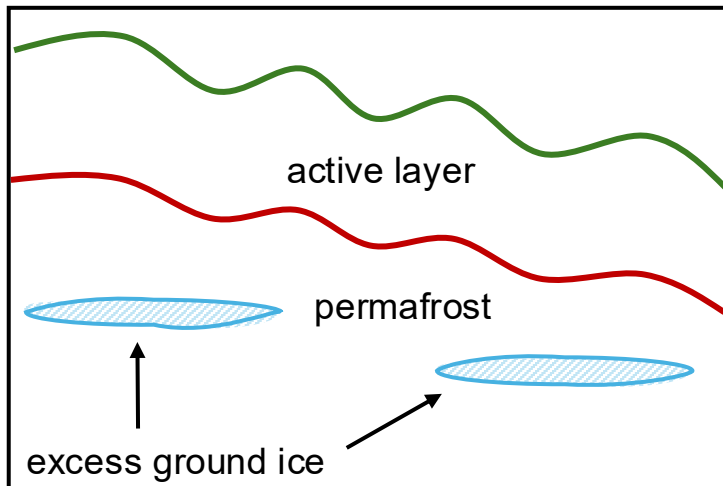


(a)

Ice-poor permafrost

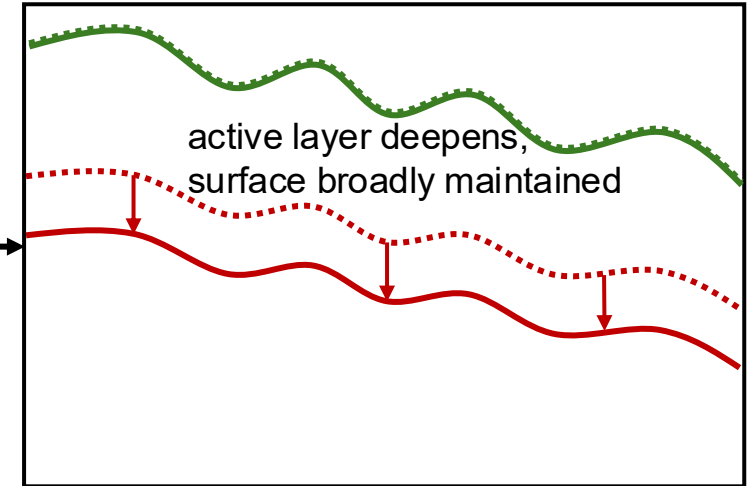


Ice-rich permafrost



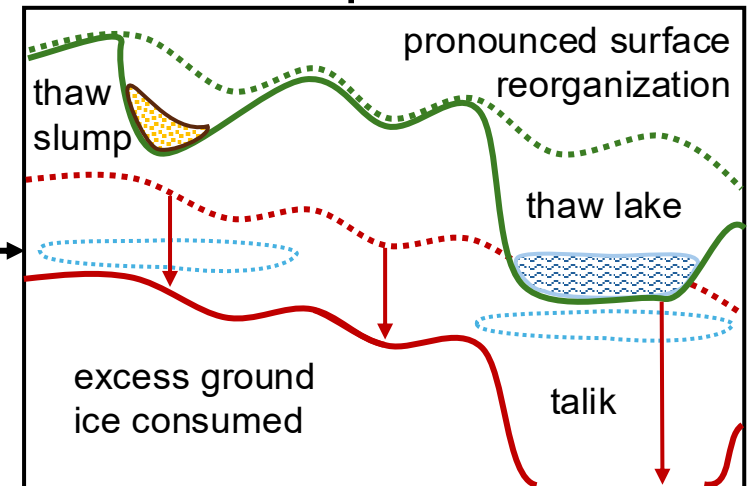
Thaw mode
focus of this study

Non-abrupt thaw



..... initial position — later position

Abrupt thaw



..... initial position — later position